

# Yelp Dataset Sentiment Analysis

Phase 2: Project Plan

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# Overview and Goals

- This project aims to predict restaurant review star ratings (1–5) from Yelp review text using text mining and machine learning.
- **Primary goal:** Build a model that predicts star ratings from review text.
- **Secondary goals:**
  - Compare feature extraction methods (n-grams, TF-IDF, embeddings).
  - Evaluate different model types (linear, tree-based, neural).
  - Address class imbalance and assess per-class performance.

*Restaurant: XYZ, Kitchener, N2G4Z6, Canada*

*Rating: ● ● ● ○ ○*

*I've been to XYZ a bunch of times. It's a decent place. Nice food, lots of variety! The place is really small though, so you almost never find a spot to sit and eat. The service is also slow at times.*

Source: Yelp Dataset Challenge: Review Rating Prediction, Nabiha Asghar

# Project Phases

# Phase 1: Data Preparation

- **Dataset:** Yelp Open Dataset (subset: specific state restaurant reviews).
- **Tasks:**
  - Filter restaurant reviews.
  - Clean and normalize text (lowercasing, punctuation removal, stopword removal).
  - Derive auxiliary features (e.g., review length, punctuation counts).
  - Split data into training, validation, and test sets. (90% training, 10% test)
- **Output:** Cleaned, balanced dataset ready for modeling.

## Phase 2: Feature Engineering

- **Approaches:**
  - Bag-of-Words with TF-IDF (unigrams, bigrams, trigrams).
  - Hashing Vectorizer for scalability.
  - Dimensionality reduction (SVD/LSI).
  - Contextual embeddings (BERT or similar).
- **Optional features:** Sentiment scores, review metadata (“useful,” “cool” votes).
- **Goal:** Identify which text representation captures rating-related patterns best.

## Phase 3: Modeling

- **Baseline models:** Logistic Regression, Naive Bayes, SVM.
- **Advanced models:** Random Forest, Gradient Boosting, Neural Networks.
- **Special approaches:**
  - Ordinal or regression-based models for ordered star ratings.
  - Ensemble or stacked models to combine strengths.
- **Goal:** Compare model families and select top performers.

## Phase 4: Evaluation

- **Validation:** k-fold cross-validation on training data.
- **Metrics:** Accuracy, RMSE, and macro-F1; per-class recall for minority ratings.
- **Analysis:**
  - Confusion matrix to diagnose class confusion.
  - Feature importance or interpretability (keywords, SHAP values).
- Test performance on held-out data.

## Phase 5: Insights and Reporting

- **Deliverables:**
  - Summary of best-performing model(s).
  - Key drivers of star ratings (important words, features).
  - Error analysis—where and why predictions fail.
- **Expected findings:**
  - Contextual embeddings and ensembles outperform simple TF-IDF models.
  - 2- and 3-star reviews remain hardest to classify due to overlapping language.



# Expected Outcomes

# Risks and Mitigations

## Risks

- Class imbalance (4-5 stars dominate)
- Overfitting on noisy text
- Ambiguity in mid range ratings
- Computational Limits

## Mitigations

- Use class weighting or resampling
- Cross-validation, regularization
- Ambiguity in mid range ratings
- Use sampling or efficient  
vectorizers

# Expected Outcomes

- Working text-to-rating prediction model achieving ~60–70% accuracy.
- Comparative evaluation of feature and model choices.
- Actionable insights into textual patterns behind customer satisfaction.